

SCUD ARSENAL

Volume 8
Issue 1



DEPARTMENT



Sri Vasavi Engineering College
Tadepalligudem

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Department Vision

To conform to prescribed academic standards and to develop socially responsible human resource in the area of Computer Science and Engineering

Department Mission

- To utilize innovative learning methods for academic improvement.
- To create a platform for learning industry-oriented technologies for better employability and self-sustenance
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character.

INTERNET OF THINGS

The Internet of things (IoT) is the inter-networking of physical devices, vehicles (also referred to as "connected devices" and "smart devices"), buildings, and other items—embedded with electronics, software, sensors, actuators, and network connectivity that enable these objects to collect and exchange data. In 2013 the Global Standards Initiative on Internet of Things (IoT-GSI) defined the IoT as "the infrastructure of the information society." The IoT allows objects to be sensed or controlled remotely across existing network infrastructure, creating opportunities for more direct integration of the physical world into computer-based systems, and resulting in improved efficiency, accuracy and economic benefit in addition to reduced human intervention. When IoT is augmented with sensors and actuators, the technology becomes an instance of the more general class of cyber-physical systems, which also encompasses technologies such as smart grids, virtual power plants, smart homes, intelligent transportation and smart cities. Each thing is uniquely identifiable through its embedded computing system but is able to interoperate within the existing Internet infrastructure. Experts estimate that the IoT will consist of almost 50 billion objects by 2020.

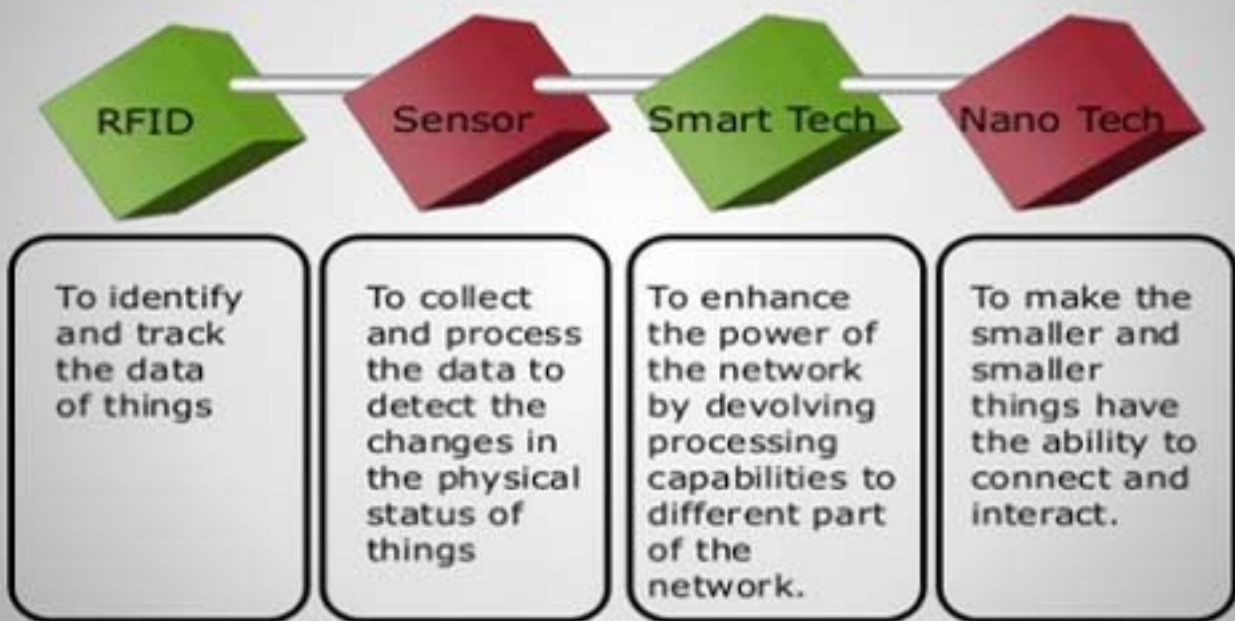
As of 2016, the vision of the Internet of Things has evolved due to a convergence of multiple technologies, including ubiquitous wireless communication, real-time analytics, machine learning, commodity sensors, and embedded systems. This means that the traditional fields of embedded systems, wireless sensor networks, control systems, automation (including home and building automation), and others all contribute to enabling the Internet of things (IoT).

IoT as a Network of Networks:



By A.Nagavalli Devi
(15A81A05C3)

How IoT Works?



The Structure of IoT

The IoT can be viewed as a gigantic network consisting of networks of devices and computers connected through a series of intermediate technologies where numerous technologies like RFIDs, wireless connections may act as enablers of this connectivity.

- **Tagging Things** : Real-time item traceability and addressability by **RFIDs**.
- **Feeling Things** : **Sensors** act as primary devices to collect data from the environment.
- **Shrinking Things** : Miniaturization and **Nanotechnology** has provoked the ability of smaller things to interact and connect within the “things” or “smart devices.”
- **Thinking Things** : **Embedded intelligence** in devices through sensors has formed the network connection to the Internet. It can make the “things” realizing the intelligent control.



Create **USD 41Billion** by providing visibility into the availability of parking spaces across the city.



Residents can identify and reserve the closest available space, traffic wardens can identify non-compliant usage, and municipalities can introduce demand-based pricing.

Efficient Waste Management in Smart Cities Supported by the Sensing-as-a-Service



[Source: "Sensing as a Service Model for Smart Cities Supported by Internet of Things", Charith Perera et. al., Transactions on Emerging Telecommunications Technology, 2014]

MICROSOFT SPEECH API

The Speech Application Programming Interface or SAPI is an API developed by Microsoft to allow the use of speech recognition and speech synthesis with in Windows applications. To date, a number of versions of the API have been released, which have shipped either as part of a Speech SDK, or as part of the Windows OS itself. Applications that use SAPI include Microsoft Office, Microsoft Agent and Microsoft Speech Server.

In general all versions of the API have been designed such that a software developer can write an application to perform speech recognition and synthesis by using a standard set of interfaces, accessible from a variety of programming languages. In addition, it is possible for a 3rd-party company to produce their own Speech Recognition and Text-To-Speech engines or adapt existing engine to work with SAPI. In principle, as long as these engines conform to the defined interfaces they can be used instead of the Microsoft-supplied engines. In general the Speech API is a freely redistributable component which can be shipped with any Windows application that wishes to use speech technology. Many versions (although not all) of the speech recognition and synthesis engines are also freely redistributable.

Basic architecture:

The Speech API can be viewed as an interface or piece of middleware which sits between applications and speech engines (recognition and synthesis). In SAPI versions 1 to 4, applications could directly communicate with engines. The API included an abstract interface definition which applications and engines conformed to. Applications could also use simplified higher-level objects rather than directly call methods on the engines. In SAPI 5 however, applications and engines do not directly communicate with each other. Instead, each talks to a runtime component (sapi.dll). There is an API implemented by this component which applications use, and another set of interfaces for engines. Typically in SAPI 5 applications issue calls through the API (for example to load a recognition grammar; start recognition or provide text to be synthesized). The sapi.dll runtime component interprets these commands and processes them, where necessary calling on the engine through the engine



interfaces (for example, the loading of a grammar from a file is done in the runtime, but then the grammar data is passed to the recognition engine to actually use in recognition). The recognition and synthesis engines also generate events while processing (for example, to indicate an utterance has been recognized or to indicate word boundaries in the synthesized speech). These pass in the reverse direction, from the engines, through the runtime dll, and on to an event sink in the application.

In addition to the actual API definition and runtime dll, other components are shipped with all versions of SAPI to make a complete Speech Software Development Kit. The following components are among those included in most versions of the Speech SDK:

- *API definition files - in MIDL and as C or C++ header files.
- *Runtime components - e.g. sapi.dll.
- *Control Panel applet - to select and configure default speech recognizer and synthesizer.
- *Text-To-Speech engines in multiple languages.
- *Speech Recognition engines in multiple languages.
- *Redistributable components to allow developers to package the engines and runtime with their application code to produce a single installable application.
- *Sample application code.
- *Sample engines - implementations of the necessary engine interfaces but with no true speech processing which could be used as a sample for those porting an engine to SAPI.
- *Documentation.

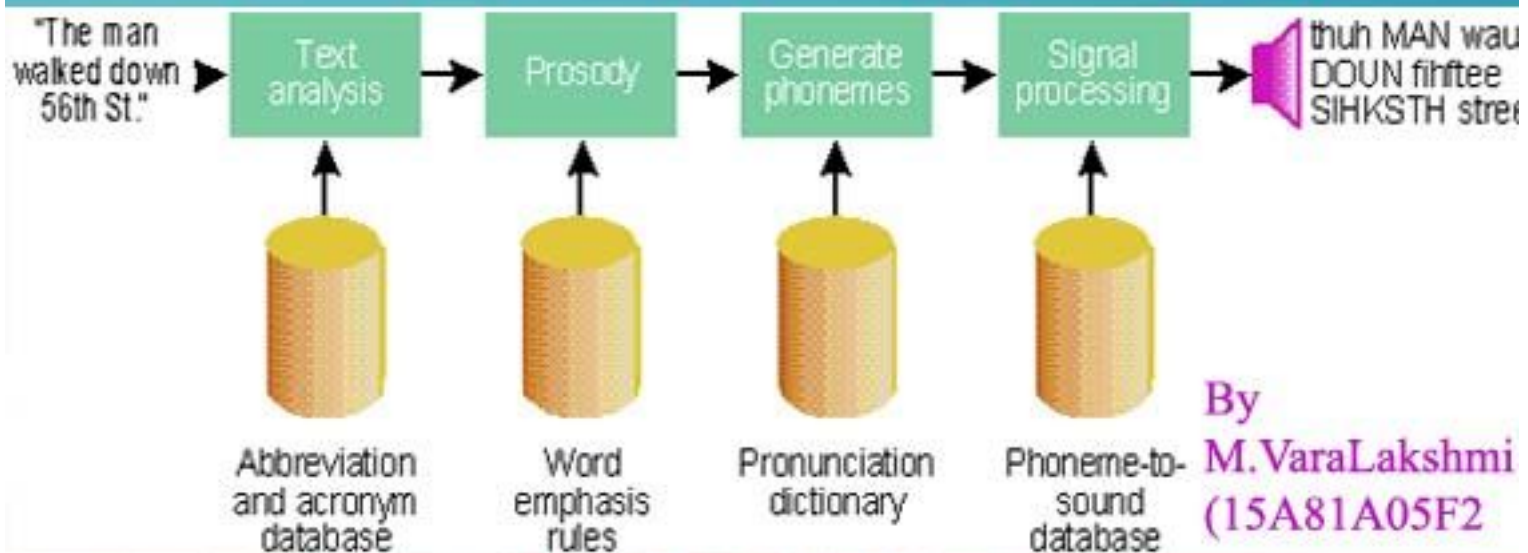
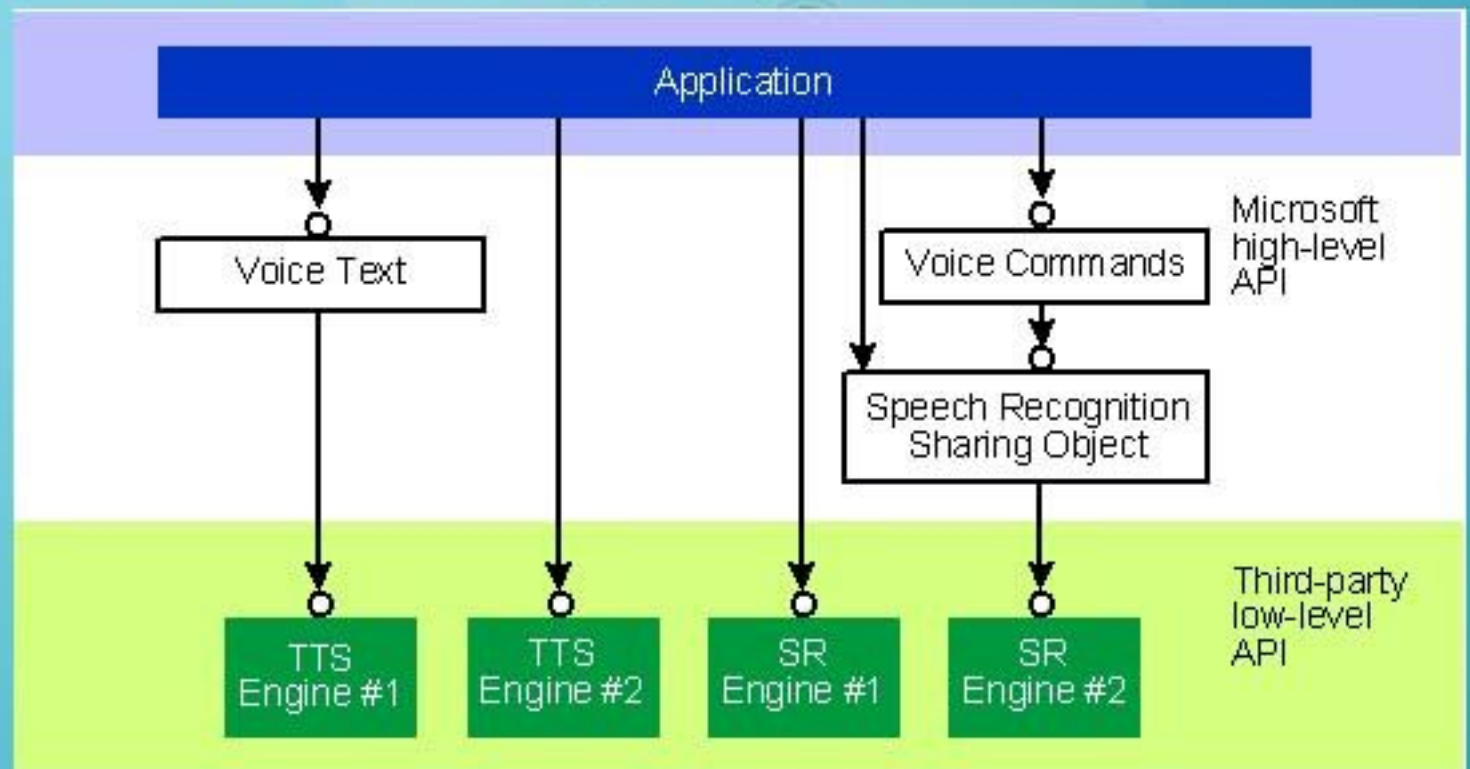
The Speech API offers two levels of access: high-level objects designed to make implementation easy, and low-level objects that offer total control but make you do a little more work. If all your program does is listen for a few voice commands and utter some simple phrases, you can use the high-level objects. To do more sophisticated stuff, you need the low-level.

The high-level objects, provided by Microsoft, don't do any SR or TTS themselves; they just call the low-level objects to do the work. The low-level objects are provided by the speech engine vendor, just like the video and



and sound card drivers that come with your display or sound card. When your app uses the low-level API, it's talking directly to the third-party code, bypassing Microsoft code completely

There are two basic technologies: speech recognition (SR) and speech synthesis, depending on who is doing the talking-you or the computer. Speech synthesis is commonly called "text-to-speech" or TTS, since the speech is usually synthesized from text data. to-speech engine.



By
M.VaraLakshmi
(15A81A05F2)

Architecture of a typical text-to-speech engine



DELPHI(Programming Language)

Embarcadero Delphi is a programming language and software development kit (SDK) for desktop, mobile, web, and console applications. Delphi's compilers use their own Object Pascal dialect of Pascal and generate native code for several platforms: Windows (x86 and x64), OS X (32-bit only), iOS (32 and 64-bit) and Android.

Delphi, part of RAD Studio, includes a code editor with Code Insight (code completion), Error Insight (real-time error-checking), and other features; refactoring; a visual forms designer for both VCL (native Windows) and FMX (cross-platform, partially native per platform) an integrated debugger for all platforms including mobile; source control (SVN, git, and Mercurial); and support for third-party plugins. It has strong database support. It is not unusual for a Delphi project of a million lines to compile in a few seconds – one benchmark gave 170,000 lines per second. It is under active development, with (in 2016) releases every six months, with new platforms being added approximately every second release.

Delphi is a strongly typed high-level programming language, intended to be easy to use and originally based on the earlier Object Pascal language. Pascal was originally developed as a general-purpose language "suitable for expressing the fundamental constructs known at the time in a concise and logical way", and "its implementation was to be efficient and competitive with existing FORTRAN compilers" but without low-level programming facilities or access to hardware. Turbo Pascal and its descendants, including Delphi, support access to hardware and low-level programming, with the facility to incorporate code written in assembly language and other languages. Delphi's object orientation features only class- and interface-based polymorphism. Metaclasses are first class objects. Objects are actually references to the objects (as in Java), which Delphi implicitly de-references, so there is usually no need to manually allocate memory for pointers to objects or use similar techniques that some other languages need. There are dedicated reference-counted string types, and also null-terminated strings.



Code examples

To show a message:

```
procedure TForm1.ShowAMessage;  
begin  
    ShowMessage('Hello World!');  
end;
```

To give a label a caption using the VCL:

```
procedure TForm1.ShowSomethingOnCreate;  
begin  
    Label1.Caption := 'Hello World!';  
end;
```

To give a label text using FireMonkey:

```
procedure TForm1.ShowSomethingOnCreate;  
begin  
    Label1.Text := 'Hello World!';  
end;
```

Programming in Delphi or any other language would not work without logic. Logic is the glue that holds together the code, and controls how it is executed.

```
var  
    number : Integer;  
    text   : String;  
begin  
    number := Sqr(17);      // Calculate the square of 17  
    if number > 400  
    then text := '17 squared > 400' // Action when if condition is true  
    else text := '17 squared <= 400'; // Action when if condition is false  
end;
```

text is set to : '17 squared <= 400'

By C.Sai Kumar
14A81A0510



Social Media Analytics and Graph Analytics

Social Media Analytics as a part of social analytics is the process of gathering data from stakeholder conversations on digital media and processing into structured insights leading to more information-driven business decisions and increased customer centrality for brands and businesses.

Types of Social Media Analytics:

Social media at a minimum has seven layers of data. Out of the seven layers, some are visible or easily identifiable (e.g., text and actions), and other are invisible (e.g., social media and hyperlink networks).

- *Textual data (such as Tweets and comments)

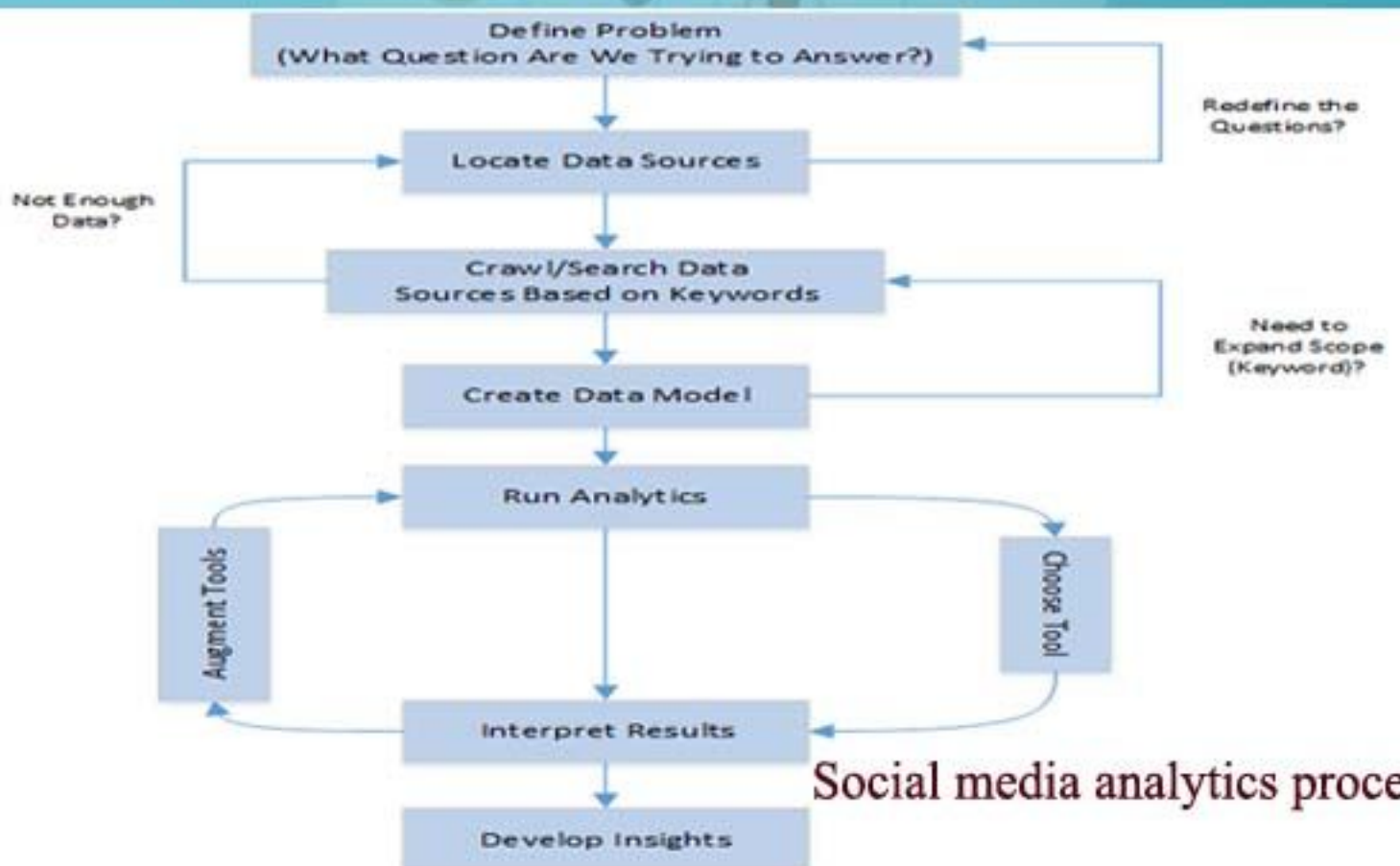
- *Network data (such as Facebook Friendship Network, and Twitter follow-following network)

- *Actions (such as likes, shares, views)

- *Hyperlinks (e.g., hyperlinks embedded within text)

- *Mobile data (e.g., mobile application data)

- *Location data



Social media analytics process

Examples of Social Data Analysis

- *Searching the web by users in order to retrieve more relevant web pages
- *Searching appropriate communities in a social network (Community Detection)
- *Search the data for queries related to individuals/groups interest and their spending pattern from shopping malls' transaction data pertaining to sales of large number of products (Recommendation & Personalization)

Graph Analytics

Graph analytics, also known as network analysis, is an exciting new area for analytics workloads. To some extent, the business driver that has shone a spot light on graph analysis is the ability to use it for social network influencer analysis.

Types of Graphs:

1. Web Graph



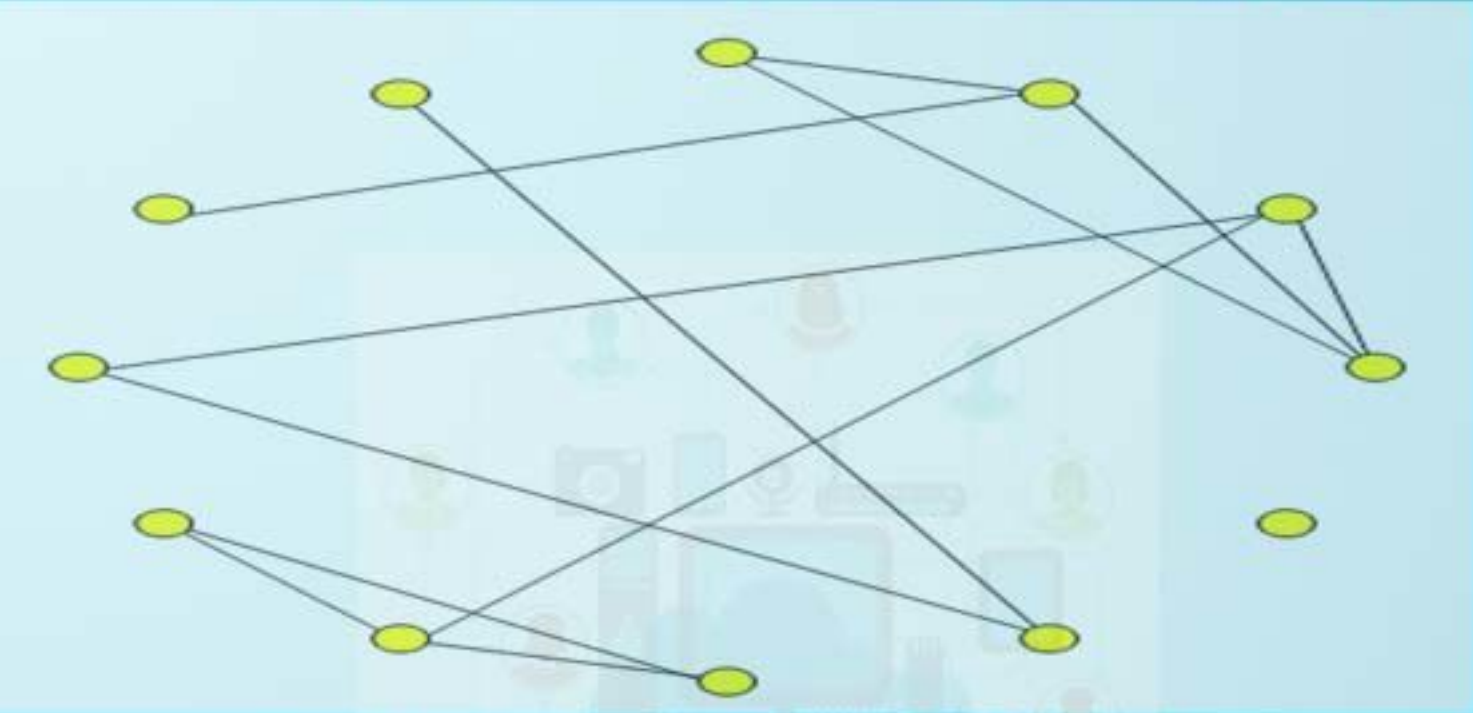
2. User-Item Graph



3. Society As a Graph



People are represented as nodes. Relationships are represented as edges.
(Relationships may be acquaintanceship, friendship, co-authorship, etc.)

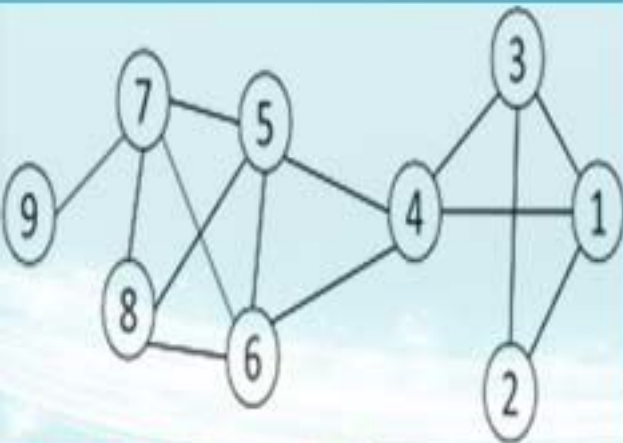


Networks and Representation

Social Network: A social structure made of nodes (individuals or organizations) and edges that connect nodes in various relationships like friendship, kinship etc.

Matrix Representation

Graph Representation



By M.Manga Raju
14A81A0537

Node	1	2	3	4	5	6	7	8	9
1	-	1	1	1	0	0	0	0	0
2	1	-	1	0	0	0	0	0	0
3	1	1	-	1	0	0	0	0	0
4	1	0	1	-	1	1	0	0	0
5	0	0	0	1	-	1	1	1	0
6	0	0	0	1	1	-	1	1	0
7	0	0	0	0	1	1	-	1	1
8	0	0	0	0	1	1	1	-	0
9	0	0	0	0	0	0	1	0	-

Details of Guest Lectures Conducted

Title of Guest Lecture	Name of Resource Person, Organization Name	Date	Target Audience	No of students attended
T & P Organized Awareness Program on Corporate Industry	Mr. D.Sudheer Kumar, Product Manager, Kurl-On, Hyderabad	04-02-2017	IV B.Tech CSE	128

Details of Faculty served as Panel Members outside

Faculty Name	Role of the faculty	Date(s)
Mr. M R Raja Ramesh	Served as Panel Member for Paper Presentation for CSI AP State Student Convention 2017 at Godavari Institute Of Engineering & Technology, Rajamahendravaram	06-01-2017 & 07-01-2017
Dr.V S Naresh	Attended BOS Meeting and Serving as Board of Studies Member, Department of Computer Science & Engineering, Pragati Engineering College, Kakinada	18-02-2017 & 19-02-2017

Placement Details

S. No	Roll No.	Name of the Student	Company
1	13A81A0560	Y.Gayathri	Virtusa Polaris
2	13A81A05C2	A.Sai Priyanka	
3	13A81A0526	Kora Janaki Durga Rathnamala	Tejas Networks
4	13A81A05A0	Pusuluri Venkata Prathyusha	
5	13A81A0569	D Sri Harshita	Avocado Global Technologies Pvt. Ltd.
6	13A81A05G4	S Satya Sirisha	



Prizes won by students in various competetions

S.NO	NAME OF THE STUDENT	EVENT, VENUE	MONTH&YEAR	PRIZE
1.	Kurmala Jayasri	Health care optimization Concept Selected for Implementation @ L&T Techgium-2016	Feb-2017	Recognition
2.	A Aditya	Paper Presentation, EMERGE 2K17, BVC College of Engineering, Rajahmundry	Feb 2017	Second Prize
3.	D Srikanth	Paper Presentation, EMERGE 2K17, BVC College of Engineering, Rajahmundry	Feb 2017	Second Prize

Various events attended by students in various colleges

S.No	Event Name	Venue	Month&Year
1.	Advanced Robotics & Embedded Systems Workshop	Sri Vasavi Engineering College	Jan-2017
2.	Cyber Security Workshop	GIET Engineering College	Jan-2017
3.	SmartIndiaHackathon	AICTE	Jan-2017
4.	National Women Parliament	Amaravathi	Feb-2017



SCUD VERVE-2K17



Various technical events and non technical events were conducted as a part of our department association function SCUD Verve 2k17.

The list of technical events is as follows:

1. Black and White Events
2. Technical Quiz
3. Technical Essay Writing
4. Code It
5. Idea Presentation

The list of non technical events is as follows:

SPORTS: (For Boys)

1. Cricket
2. Volley Ball
3. Badminton
4. Chess
5. Kabaddi

SPORTS: (For Girls)

1. Kho Kho
2. Throw Ball
3. Badminton
4. Chess
5. Carroms

Others:

6. Rangoli



Prizes won by students in Technical events

BLACK & WHITE EVENT:

Prize	Roll Number	Student Name
First	13A81A0544	PINNAMARAJU PRASANTHI
Second	15A81A0528	MAJETY H V A SRI VYSHNAVI

TECHNICAL - QUIZ:

Prize	Roll Number	Student Name
First	14A81A0531	K.VIJAYALAKSHMI
	14A81A0535	K.PADMINI ISHAWARYA
Second	15A81F0005	K.ASWINI
	15A81F0008	Y.RAMYA
	15A81F0007	U.DEVI DIVYA

TECHNICAL ESSAY WRITING:

Prize	Roll Number	Student Name
First	15A81A0540	RALLAPALLI MOUNICA
Second	14A81A05C7	BOLLISSETTI ALEKHYA

CODE IT:

Prize	Roll Number	Student Name
First	14A81A0577	KALLA RAJAGANGADHAR
Second	15A81A0586	MADUPALLI JAYANTH

IDEA PRESENTATION:

Prize	Roll Number	Student Name	Title
First	14A81A0504	AMBATI N V D SAI ADITYA	FIREBASE
Second	15A81A05L4	MOHAMMAD YASMIN	EYEGAZE COMMUNICATION SYSTEM

Merit Scholarships

Merit Scholarships were given to the students who secured below 20000 rank in Eamcet and also who performed well in University end examinations. Details of the scholarship list are as follows:

S.NO	Eamcet Rank	Roll No	Name of the Student	Class	Amount Recieved
1.	12339	13A81A0560	YERRA GAYATHRI	IV Year-A	20000
2.	14504	13A81A05A0	PUSULURI VENKATA PRATHYUSHA	IV Year-B	20000
3.	18927	13A81A05H2	YERRA BHAVYA	IV Year-C	15000
4.	22845	15A81A0505	CHALAMALASETTI JHANDINI	II Year-A	10000
5.	13206	15A81A0510	GARLAPATI NAVYA	II Year-A	20000
6.	22595	15A81A0514	K DEEPIKA	II Year-A	10000
7.	14476	15A81A0525	KOTHA NALINI VENKATA LAKSHMI	II Year-A	20000
8.	9401	15A81A0526	KOTHA SWATHI SRI SAI NAGA DURGA	II Year-A	20000
9.	24407	15A81A0527	LINGALA INDIRA PRAVALLIKA	II Year-A	10000
10.	17580	15A81A0528	MAJETY H V A SRI VYSHNAVI	II Year-A	15000
11.	20750	15A81A0531	MATHI LIKHITHA	II Year-A	10000
12.	18501	15A81A0546	THOTA NEERAJA	II Year-A	15000
13.	12753	15A81A0547	TUMMALAPALLI SUPRIYA	II Year-A	20000
14.	23419	15A81A0550	VANNEMREDDY RAMALAKSHMI	II Year-A	10000
15.	24627	15A81A0552	VARANASI LAKSHMIMEGHANA	II Year-A	10000
16.	19964	15A81A0562	AINAMPUDI LAVANYA DEVI	II Year-B	15000
17.	18007	15A81A0576	KANDUKURI AKHILA	II Year-B	15000
18.	16953	15A81A0589	MARAGANI SRAVANI DURGA	II Year-B	15000
19.	22169	15A81A0595	NOWBATHULA DURGAPRIYA	II Year-B	10000
20.	10974	15A81A0597	PAILU KAVYA PRIYA	II Year-B	20000
21.	12987	15A81A05A2	PENDYALA DURGA SUNEETHA	II Year-B	20000

S.NO	Eamcet Rank	Roll No	Name of the Student	Class	Amount Recieved
22.	23126	15A81A05B2	SRIRAM VENKATA SUBBA RAO	II Year-B	10000
23.	17832	15A81A05B7	VADREVV SIVA SANKAR	II Year-B	15000
24.	22780	15A81A05B8	VALLIPALLI HANUMA NAGA SUSMITHA	II Year-B	10000
25.	14813	15A81A05C0	VUDDAGIRI INDUMATHI	II Year-B	20000
26.	16615	15A81A05C3	ALLA NAGAVALLI DEVI	II Year-C	15000
27.	18035	15A81A05C4	ALLAM SNEHA SRI SANDHYA	II Year-C	15000
28.	24783	15A81A05D1	CHILUKURI LAKSHMI CHANDINI	II Year-C	10000
29.	22794	15A81A05E1	G DHATRI SRI VINAYA	II Year-C	10000
30.	15543	15A81A05E4	KOLLURI MANIKUMARI	II Year-C	15000
31.	24524	15A81A05E6	K SAI LAKSHMI PRASANNA	II Year-C	10000
32.	12854	15A81A05F1	MADDAMSETTI LAKSHMI MOUNICA	II Year-C	20000
33.	17187	15A81A05F5	MUSUNURU BALA KRISHNA	II Year-C	15000
34.	23829	15A81A05F6	NADIMPALLI HEMALAKSHMI BHAVANI	II Year-C	10000
35.	19953	15A81A05F8	NEKKANTI VYSHNAVI	II Year-C	15000
36.	10989	15A81A05H0	SEELAM SOWJANYA	II Year-C	20000
37.	15865	15A81A05I1	ADDANKI DIVYA	II Year-D	15000
38.	12874	15A81A05I5	ARIMILLI SRINIVASA SANTOSH KUMAR	II Year-D	20000
39.	12730	15A81A05K0	KARRI CHIRANJEEVI SAI ISWARYA	II Year-D	20000
40.	21703	15A81A05M0	NARRA MOUNIKA	II Year-D	10000
41.	16230	15A81A05M6	POSINA SASANKA	II Year-D	15000
42.	23478	15A81A05M7	PULAPA SAI RAMYA	II Year-D	10000
43.	11709	15A81A05N2	SUGGISETTI SOWMYA SRI	II Year-D	20000
44.	23032	15A81A05N8	VASA CHAITANYA	II Year-D	10000
45.	11036	16A81A0507	CH LAKSHMI BHAVANI	I Year-A	15000

S.NO	Eamcet Rank	Roll No	Name of the Student	Class	Amount Recieved
46.	4258	16A81A0535	MOKA LAKSHMI SAILAJA	I Year-A	20000
47.	13139	16A81A0541	PEDAPUDI PRANEELA	I Year-A	15000
48.	10863	16A81A0544	P SURYAKANTH	I Year-A	15000
49.	13913	16A81A0571	BANDI SAI JAYANTH	I Year-B	15000
50.	9828	16A81A0590	JAMPANA SAILAJA	I Year-B	20000
51.	15019	16A81A05A0	MEDURI HIMASRI	I Year-B	10000
52.	11267	16A81A05A1	MEDURI SRI MAHI	I Year-B	15000
53.	11493	16A81A05A3	MYLAVARAPU SONIKA	I Year-B	15000
54.	12736	16A81A05G5	P VENKATA MOUNIKA	I Year-C	15000
55.	9442	16A81A05H9	VELAGALA MEGHANA	I Year-C	20000
56.	9927	16A81A05N5	V MAHALAKSHMI	I Year-D	20000



SKUD VERVE 2K17



FAREWELL DAY CELEBRATIONS

